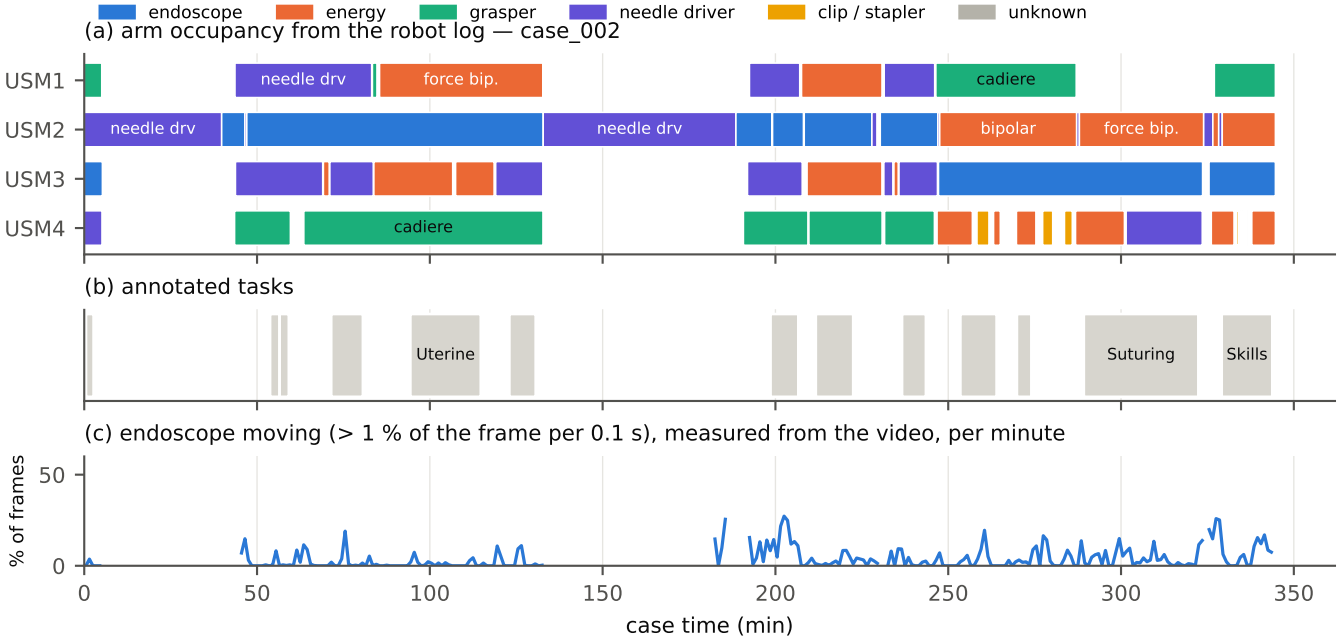
endoscope
energy
grasper
needle driver
clip / stapler
unknown
(a) arm occupancy from the robot log — case_002
USM1
USM2
USM3
USM4
needle drv
force bip.
cadiere
needle drv
needle drv
bipolar
force bip.
needle drv
cadiere
(b) annotated tasks
Uterine
Suturing
Skills
(c) endoscope moving (> 1 % of the frame per 0.1 s), measured from the video, per minute
% of frames
50
0
0
50
100
150
200
250
300
350
case time (min)